



Loop a Pattern in Code

Ontario Math C3.1 — repeating events

Name: _____

Date: _____

★ I can loop a small pattern of moves to make a staircase.

Repeat a little pattern

A staircase is the same step over and over. Bit's step is: go across, then go up. A loop repeats that step to build the whole staircase.

Read the code — `range(3)` repeats the across-then-up step three times.

Read this Python code

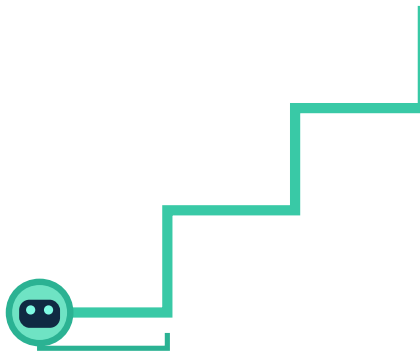
PYTHON

```
for step in range(3):  
    bit.forward(30)  
    bit.left(90)  
    bit.forward(30)  
    bit.right(90)
```

OUTPUT

Bit climbs a 3-step staircase.

`range(3)` → 3 steps up



Each repeat is one step: across, then up. `range(3)` makes 3 steps.

1 Read it

How many steps does the staircase have? _____ Each step has 2 forward moves, so how many forward moves in TOTAL? _____

2 Predict, then run

You change range(3) to range(5). PREDICT: how many steps, and how many forward moves in total? Then run it.

3 Write your own

Make Bit climb a 4-step staircase. Write the range number: for step in range(_____):

4 Challenge

Each step goes UP 30. If Bit climbs a 4-step staircase, how high does Bit go in total? Show your thinking.

Big idea

A loop can repeat a little PATTERN of moves, not just one move. Repeat the across-then-up step and Bit builds a whole staircase.